

Something

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Algorithm 1: Test statistic ϕ (Kuditipudi et al.)

Input : string $y \in \mathcal{V}^*$, watermark key sequence $\xi \in \Xi^n$
Params: alignment cost d , block size k
Output: test statistic value $\phi(y, \xi) \in \mathbb{R}$

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1 for  $i \in 1, \dots, \text{len}(y) - k + 1$  do
2   for  $j \in 1, \dots, n$  do
3      $\mathbf{y}^i \leftarrow \{y_{i+\ell}\}_{\ell=0}^{k-1}$ ,  $\xi^j \leftarrow \{\xi_{(j+\ell)\%n}\}_{\ell=0}^{k-1}$ ;
4      $\hat{d}_{i,j} \leftarrow d(\mathbf{y}^i, \xi^j)$ ;
5   end
6 end
7 return  $\min_{i,j} \hat{d}_{i,j}$ ;
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Because of how the g_ρ are chosen, we will be done if we can show, for any particular y_{i_1} , ξ_{i_2} , and g_{i_3} , that $g_{i_3}(y_{i_1}, \xi_{i_2})$ shows up in only one of the $\{d(y^i, \xi^j)\}$. In other words, we must show that, once y_{i_1} , ξ_{i_2} , and g_{i_3} are picked, i and j of $d(y^i, \xi^j)$ are uniquely determined. $\square$

TODO, proof isn't true.

1 Something

From SynthID-text, we have

$$\text{Score}(x) = \frac{1}{mT} \sum_{t=1}^T \sum_{\rho=1}^m g_\rho(x_t, r_\rho).$$

Define

$$\text{Score}(x, \xi) = \frac{1}{mT} \sum_{t=1}^T \sum_{\rho=1}^m g_\rho(x_t, \xi_\rho) \quad (1)$$

and let

$$d(\cdot, \cdot) = -\text{Score}(\cdot, \cdot). \quad (2)$$

Theorem 1.1. *Let ϕ be the output of Algorithm 1 when ran on an unwatermarked string $y \in V^*$, watermark key $\xi \in \Xi^n$, alignment function as defined in Equation 2 and chunk size k . Let F_ϕ and f_ϕ be the distribution and density of ϕ respectively. Then*

$$F_\phi(x) = 1 - (1 - F_{\text{Score}}(x))^\mu$$

$$f_\phi(x) = \mu f_{\text{Score}}(x) (1 - F_{\text{Score}}(x))^{\mu-1}$$

where $\mu = (\text{len}(y) - k + 1) \cdot n$ and $F_{\text{Score}}/f_{\text{Score}}$ are the distributions and densities of $\text{Score}(y, \xi)$ for y unwatermarked and $\xi \sim \nu$ (in the $N = 2$ case, $\text{Score}(y, \xi) \sim \text{Binomial}(m \cdot T, 0.5)$).

Proof. Observe that the distribution of ϕ in the theorem is the distribution of the minimum of m i.i.d. random variables with density f_{Score} and distribution F_{Score} ; similarly for the density. It suffices to show then that $\{d(y^i, \xi^j) : i \in 1, \dots, \text{len}(y) - k + 1, j \in 1, \dots, n\}$ are i.i.d.

To show this, recall from Equation 1 that

$$d(y^i, \xi^j) = -\frac{1}{mT} \sum_{t=1}^k \sum_{\rho=1}^m g_\rho(y_t^i, \xi_\rho^j). \quad (3)$$